

Space Complexity : Input space + Auxiliary space

Input space : It is the space required for the input data on which we are going to perform the operation.

Auxiliary space : It is the extra space required for the variables to perform that operation.

Input space --> `Int arr[9]` :
Input space is $O(n)$
Here, n is the number of elements in the array.

Aux space : key , loop variable
Irrespective of the input space, the auxiliary space is constant.
Eg : for 900 array elements, we still need 1 key variable and 1 loop variable.
If the auxiliary space is constant, we say that it is $O(1)$

Total space complexity for linear search :
Input + aux :
 $O(n) + O(1)$
 $O(n+1)$
Total space complexity : $O(n)$ --> $+1$ is discarded as per the theory of probability.

Space complexity :
Only Auxiliary space is considered : $O(1)$